

Tree water sources over shallow, saline groundwater in the lower River Murray, south-eastern Australia: implications for groundwater recharge mechanisms

K. L. Holland^{A,D}, S. D. Tyerman^B, L. J. Mensforth^C and G. R. Walker^A

^ACSIRO Land and Water, PMB 2, Glen Osmond, SA 5064, Australia.

^BUniversity of Adelaide, School of Agriculture and Wine, PMB 1, Glen Osmond, SA 5064, Australia.

^CDepartment of Water, Land and Biodiversity Conservation, GPO Box 2834, Adelaide, SA 5001, Australia.

^DCorresponding author. Email: kate.holland@csiro.au

Abstract. The decline of riparian vegetation in the lower River Murray, south-eastern Australia, is associated with a reduction in flooding frequency, extent and duration, and increased salt accumulation. The plant water sources of healthy *Eucalyptus largiflorens* trees growing over highly saline ($>40 \text{ dS m}^{-1}$) groundwater were investigated during summer when water deficit is greatest. The study found low-salinity soil water overlying highly saline groundwater at most sites. This deep soil water, rather than the saline groundwater, was identified as the plant water source at most sites. Stable isotopes of water and water potential measurements were used to infer how the deep soil water was recharged. The low-salinity, deep soil water was recharged in the following two ways: (1) vertically through the soil profile or via preferential flow paths by rainfall or flood waters or (2) horizontally by bank recharge from surface water on top of the saline groundwater. Vertical infiltration of rainfall and floodwaters through cracking clays was important for trees growing in small depressions, whereas infiltration of rainfall through sandy soils was important for trees growing at the break of slope. Bank recharge was important for trees growing within $\sim 50 \text{ m}$ of permanent and ephemeral water bodies. The study has provided a better understanding of the spatial patterns of recharge at a scale relevant to riparian vegetation. This understanding is important for the management of floodplain vegetation growing in a saline, semi-arid environment.

Introduction

In regions that experience temporal water deficits, the distribution of terrestrial plants is most often regulated by the availability of water during critical periods (Kozłowski 1968). The decline of riparian vegetation in the lower River Murray, south-eastern Australia has been linked to a reduction in flooding frequency, extent and duration, and increased salt accumulation (Murray–Darling Basin Ministerial Council, MDBMC 1999; Slavich *et al.* 1999). It is estimated that 40% of the floodplain vegetation is currently affected by salinisation, comprising vegetation communities dominated by unhealthy or dead trees and halophytes (Department for Environment and Heritage, DEH 2004). To better manage these floodplains, we need to understand the processes controlling long-term plant water availability and vegetation health (Eagleson 1982; Nemani and Running 1989; Hatton and Evans 1998).

Stable isotope techniques have been used in conjunction with water-potential measurements to determine from where plants were obtaining their water. The plant water sources that have been identified by using these techniques include

hydraulic redistribution, where trees transport water from the groundwater into the surface soils (Dawson 1993; Dawson and Pate 1996; Burgess *et al.* 1998; Caldwell *et al.* 1998), and from the surface soils, following rainfall into the deeper root zone (Burgess *et al.* 2001). Isotopic methods have also been used to differentiate plant water sources spatially and temporally, where riparian trees have access to water from creeks, soil and groundwater (White *et al.* 1985; Dawson and Ehleringer 1991; Flanagan *et al.* 1992; Thorburn *et al.* 1993a; Mensforth *et al.* 1994; Dawson and Pate 1996; Mensforth and Walker 1996; Chimner and Cooper 2004).

The floodplains of the lower River Murray are typically vegetated by a mixture of river red gum (*Eucalyptus camaldulensis*), black box (*E. largiflorens*) and lignum (*Muehlenbeckia florulenta*) (O'Malley and Sheldon 1990). River red gums cover $\sim 20\%$ of the vegetated area (DEH 2004) and tend to grow in less saline, more frequently flooded parts of the floodplain, typically adjacent to creek beds. Whereas black box cover $\sim 30\%$ of the vegetated area (DEH 2004) and are found at higher elevations away from the creeks, but with access to shallow groundwater

(Slavich *et al.* 1999). Lignum cover $\sim 15\%$ of the vegetated area (DEH 2004) and grow in the lower, more frequently flooded areas of the floodplain (Craig *et al.* 1991).

Detailed investigations of black box plant water sources found that they were reliant on moderately saline groundwater ($<40 \text{ dS m}^{-1}$), and occasionally water that had infiltrated from floods or rainfall (Thorburn *et al.* 1993a). However, many parts of the lower River Murray floodplain have healthy black box trees growing over highly saline ($>40 \text{ dS m}^{-1}$) groundwater that is not available for plant water uptake. This suggests that these trees had access to alternative plant water sources to survive extended periods of water deficit that are typical of these semi-arid floodplains.

We hypothesised that trees growing over highly saline ($>40 \text{ dS m}^{-1}$) groundwater had access to lower-salinity water sources, including soil water in the unsaturated zone and surface water (Fig. 1). In this paper we test this hypothesis by using stable isotopes of water and water-potential techniques to identify the plant water sources of black box trees growing over highly saline groundwater during a period of water deficit (i.e. summer). Tree locations were selected to test the effect that variations in floodplain topography, distance from surface water and groundwater salinity had on plant water sources. We then used the plant water-source results to infer

the recharge mechanisms that maintain these alternative plant water sources during periods of water deficit.

Materials and methods

Site description

Field investigations were conducted on three floodplains in the lower River Murray, Clarks and Gurra floodplains (near Loxton; $34^{\circ}44'S$, $140^{\circ}60'E$) and Yarra floodplain (near Waikerie; $34^{\circ}18'S$, $139^{\circ}98'E$; Fig. 2). The climate in the lower River Murray is semi-arid with mild winters and long hot summers. Mean annual potential evaporation ($1830 \text{ mm year}^{-1}$) is greater than seven times mean annual rainfall ($\sim 275 \text{ mm year}^{-1}$). Annual rainfall is highly variable, with recorded values at Loxton ranging from 125 mm year^{-1} up to 475 mm year^{-1} and includes maximum daily rainfall values of between 24 mm day^{-1} and 68 mm day^{-1} . Annual rainfall was below average during the field investigations in 2001 (250.2 mm at Loxton and 229.9 mm at Waikerie; Bureau of Meteorology 2001). River flows were at base flow ($5\text{--}10 \text{ GL day}^{-1}$) during the sampling period following a minor flood (63.5 GL day^{-1}) in December 2000 (Murray–Darling Basin Commission, MDBC 2001) that did not inundate any of the sites.

Samples of soil water from the unsaturated zone, rainwater, surface water from the River Murray, nearby creeks and lakes and groundwater from site piezometers were collected in February 2001 (late summer). Samples were analysed for stable isotopes of water and water potentials as described below. Water potential and stable isotope of water measurements were used to identify plant water sources of 12 trees from eight sites. Soil profiles were taken at two sites (CF12, CF14) on Clarks floodplain, five sites (GG2, GG5, GG6, GG8, GG9) on Gurra floodplain and one site (Y1) on Yarra floodplain.

Plant and soil measurements

The isotopic composition ($\delta^2\text{H}$ and $\delta^{18}\text{O}$) of water in soil and plant samples was measured, along with soil and twig water potentials. Field

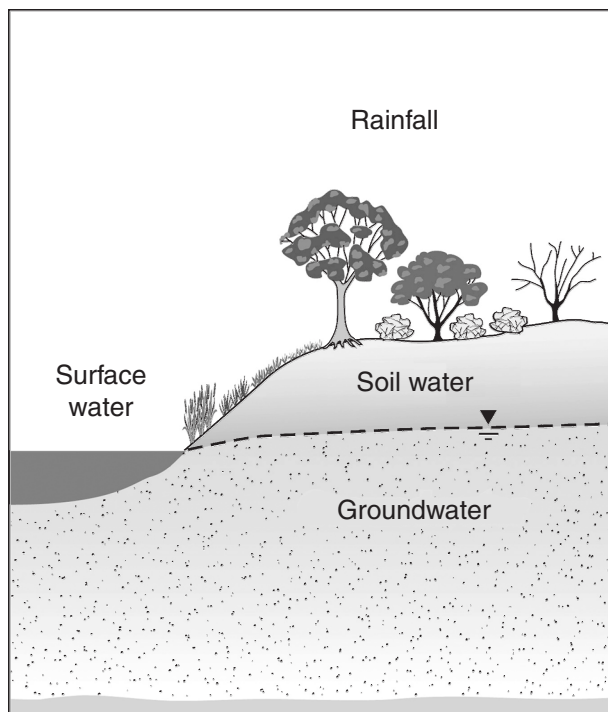


Fig. 1. Potential plant water sources on the lower River Murray floodplain investigated in this paper. Potential plant water sources that are available to black box trees during periods of water deficit between floods include soil water stored in the soil profile above the groundwater, the groundwater and surface water from the River Murray, ephemeral creeks or lakes.

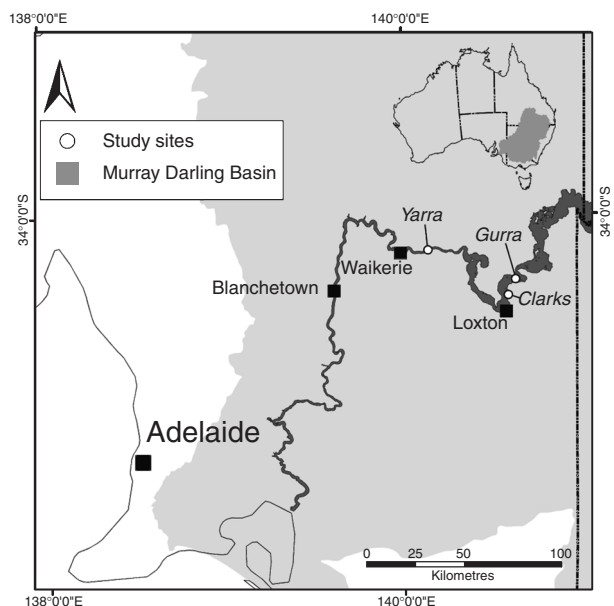


Fig. 2. Study area showing the location of the three floodplains in the lower River Murray, Clarks and Gurra floodplains (near Loxton; $34^{\circ}44'S$, $140^{\circ}60'E$) and the Yarra floodplain (near Waikerie; $34^{\circ}18'S$, $139^{\circ}98'E$).

samples were taken at the end of summer (13–15 February 2001) during the period of greatest annual plant water stress. Light rain (12 mm at Loxton) fell on 13 February 2001, prior to the sampling trip, potentially reducing plant water stress.

Leaf water potential (Ψ , MPa) was measured on three replicate twigs per tree taken 1–2 m from the ground before dawn at all sites with a Scholander pressure bomb (Scholander *et al.* 1965; Ritchie and Hinckley 1975). Leaves were bagged and placed in a humidified bomb chamber to minimise water loss during measurement (Turner 1988). Leaf water potentials at predawn were considered to be equivalent to stem water potentials. Leaf water potential values were compared with soil water potential values to determine which parts of the profile were available to the tree.

Soil water potential (Ψ , MPa) values represent the sum of the matric (dryness) and osmotic (saltiness) water potentials, where a water potential of 0 MPa is equivalent to free water and more negative values are drier and/or more saline. Water must move down the hydraulic gradient from soil to root (Gardner 1960), therefore only soil regions with higher (less negative) soil water potential than the root water potential are available to the tree at any given time. This means that soil water potential values that are less negative than predawn leaf water potentials (when stem water potentials are at a maximum) indicate regions in the soil from which the roots are able to extract water. Previous work on black box trees in the lower River Murray indicated that they were in water potential equilibrium with these regions of the soil at predawn (Zubrinich *et al.* 2000). Twig water samples from each tree were taken and analysed for stable isotopes of water, following the method described by Thorburn *et al.* (1993b), who showed that young *E. largiflorens* do not fractionate stable isotopes of water during transport in the xylem to the sampled twigs.

Soil profiles were taken from eight sites and sampled at 0.2-m increments from the surface down to 1.0-m depth, and at 0.5-m depth intervals from 1.0 m to the saturated zone (typically at 2–4-m depth). Soil samples were analysed for matric potential, soil water chloride concentration and stable isotopes of water. Matric potential (Ψ_m , MPa) was determined by the 'filter paper' technique (Greacen *et al.* 1989). Total chloride was measured colorimetrically in water extracts (Taras *et al.* 1975), then converted to the chloride concentration in the soil solution (mg L^{-1}) by using the gravimetric water content. Gravimetric water content (g g^{-1}) was measured by oven drying at 105°C for 24 h. Osmotic potential (Ψ_π , MPa) was estimated from the chloride concentration of the soil solution ($[\text{Cl}^-]$, mg L^{-1}), calculated by using the Van't Hoff equation and an osmolality of $1.848 \text{ Os kg water}^{-1}$ for a 0.995 M NaCl solution at 20°C (Weast and Astle 1980), which has an osmotic potential of 4.524 MPa. This relationship was used to calculate the effective osmotic potential of the soil-water chloride concentration as follows:

$$\Psi_\pi = [\text{Cl}^-] \times 4.524/35.45. \quad (1)$$

This method assumes that all salts in the soil solution were present as NaCl and that the concentration used to calculate this relationship is appropriate for the range of soil salinities encountered by black box trees. Total soil water potential (Ψ , MPa) was obtained by summing the matric and osmotic potentials. Gravitational water potentials (0.01 MPa m^{-1}) were not included in this value as they are small in comparison with measured matric and osmotic potentials.

Groundwater depth (m) was measured in the piezometer at each site and used to calculate the groundwater depth below each tree by using surveyed elevations. Piezometers were installed 3–5 m below the water table, but sometimes up to 15 m below the water table (e.g. site Y1). Groundwater electrical conductivity (dS m^{-1}) was measured following purging of the well and prior to sampling for isotopes. The groundwater electrical conductivity (dS m^{-1}) was

converted to a chloride concentration ($\text{mg Cl}^- \text{ L}^{-1}$) assuming that all salts were present as NaCl, a reasonable assumption in the Murray Basin (Herczeg *et al.* 2001). The groundwater chloride concentration was converted to groundwater osmotic potential as described for soil chloride concentrations above. Osmotic potentials were considered to equate to groundwater potentials at these sites as groundwater was shallow (<5 m) and gravitational water potentials are small in comparison with measured osmotic potentials.

Isotope measurements

The plant water-sourcing method using stable isotopes depends on different sources of water having different isotopic values. These differences can arise owing to transport processes (Gat 1981) and phase transitions through the atmosphere, lithosphere and biosphere (Barnes and Allison 1983). The method is based on the assumption that the isotopic composition of water is not altered when (1) water is taken up by roots and (2) when it moves from roots to leaves (Thorburn *et al.* 1993b). Plant water sources were determined by comparing the twig isotope value with that of the possible source waters that were within the range of water-potential values available to the plant.

Samples of rainfall from Clarks floodplain (taken on 13 February 2001), river water, lake water and groundwater were taken from each site. Water samples were extracted from soil and plant samples by azeotropic distillation with kerosene, following Revesz and Woods (1990) and Thorburn *et al.* (1993b), respectively. Water samples ($25 \mu\text{L}$) were vaporised and circulated over uranium metal heated to 800°C to determine $\delta^2\text{H}$ values (Dighton *et al.* 1997). $\delta^{18}\text{O}$ water samples were equilibrated with CO_2 , following Socki *et al.* (1992). Samples were analysed with a dual-inlet gas-ratio mass spectrometer (Europa Scientific Ltd). Isotopic concentrations were expressed as delta (δ) values per mil (‰) relative to the international standard, V-SMOW (Vienna-Standard Mean Ocean Water). Delta values were calculated by using the following formula:

$$\delta = [R_i/(R_s - 1)] \times 1000, \quad (2)$$

where R_i is the ratio of the heavy to the light isotope in a sample and R_s the same ratio in the standard. The error on plant and soil water analysis is $\pm 1.3\%$ for $\delta^2\text{H}$ and $\pm 0.3\%$ for $\delta^{18}\text{O}$ (Thorburn *et al.* 1993b). Plotting the $\delta^2\text{H}$ values of possible water sources and twig water samples against their corresponding $\delta^{18}\text{O}$ values was used to identify plant water sources. Plant water sources were further constrained by excluding those samples where total soil water potential was less than -3.0 MPa , following Thorburn *et al.* (1993b). This corresponds to the minimum soil water potential for extraction of water by *E. largiflorens* (Slavich *et al.* 1999) and measured predawn water potentials.

Results

Plant water sources

Surface water samples from Gurra Lake (-16.5% $\delta^2\text{H}$ and -4.44% $\delta^{18}\text{O}$) and the creek near GG6 (3.2% $\delta^2\text{H}$ and 2.78% $\delta^{18}\text{O}$) were isotopically enriched relative to River Murray water samples taken near CF12 and Y1 (Table 1, Fig. 3), and the local meteoric water line $\delta^2\text{H} = 7.6 \delta^{18}\text{O} + 8$ (Simpson and Herczeg 1991). The differences in evaporative enrichment of the backwater samples relative to River Murray samples appeared to be related to residence times and surface water area. Groundwater samples were isotopically depleted relative to the soil and surface water samples, ranging from -23.1% $\delta^2\text{H}$

Table 1. Groundwater depth and salinity, water potential (Ψ) and isotope values of twig and water samples taken from Clarks, Gurra and Yarra floodplains in February 2001
PD, piezometer depth below the natural surface

Tree			Groundwater				Deep soil water			Surface water		Twig water		
	Depth (m)	PD (m)	Salinity (dS m ⁻¹)	Ψ (MPa)	$\delta^2\text{H}$ (‰)	$\delta^{18}\text{O}$ (‰)	Ψ (MPa)	$\delta^2\text{H}$ (‰)	$\delta^{18}\text{O}$ (‰)	$\delta^2\text{H}$ (‰)	$\delta^{18}\text{O}$ (‰)	Ψ (MPa)	$\delta^2\text{H}$ (‰)	$\delta^{18}\text{O}$ (‰)
CF12a	3.86													
CF12b	3.51	4.1	1.15	-0.06	-33.6	-4.32	-0.66	-23.7	-3.09	-6.0	0.04	-1.08	-24.5	-3.20
CF14c	2.24	3.4	73.10	-3.61	-31.6	-4.57	-1.63	-31.9	-3.42	—	—	-1.85	-26.3	-2.99
GG2a	1.54	10.0	73.70	-3.64	-28.1	-3.70	-1.55	-10.6	-0.19	16.5	4.44	-1.80	-28.0	-2.52
GG5a	0.99	10.0	52.60	-2.59	-23.1	-2.26	-0.95	23.3	6.06	16.5	4.44	-1.53	-8.9	0.73
GG5b	1.29						-0.98	-20.1	-1.63			-1.73	18.2	6.03
GG6a	3.80	10.0	67.00	-3.30	-26.1	-2.57	-1.18	-12.3	-0.70	3.2	2.78	-2.55	-18.8	-0.69
GG8a	1.13											-1.93	-13.0	-0.42
GG8c	2.85	10.0	71.10	-3.51	-31.3	-4.04	-0.59	1.4	1.96	16.5	4.44	-1.38	-5.2	1.40
GG9a	1.24	10.0	95.80	-4.73	-30.9	-3.74	-2.31	-27.8	-3.34	16.5	4.44	-1.38	-3.4	2.10
Y1a	4.03											-2.20	-25.8	-2.51
Y1b	4.13	18.6	31.60	-1.56	-37.6	-4.61	-2.06	-32.0	-3.83	-6.1	-0.29	-1.78	-12.3	-0.21
												-2.55	-12.4	0.44

and -2.26‰ $\delta^{18}\text{O}$ at GG5 to -37.6‰ $\delta^2\text{H}$ and -4.61‰ $\delta^{18}\text{O}$, respectively, at Y1 (Table 1).

Soil profiles were characterised by relatively high soil water potentials in the deepest soil samples (greater than -2.31 MPa; Table 2), meaning that this water was available to the trees. The water potentials of the groundwater were more negative than for the deepest soil water samples, except at site CF12a. Soil water potentials were more negative than predawn water potentials, except within 0.5–1.0 m of the water table at sites CF12a, CF12b, CF14c, GG2a, GG5b, GG6a, GG8a, GG8c and GG9a. Plant roots are more likely to extract water from zones in the soil where the soil water potential is less than or equal to the predawn water potential. Since during the night, trees equilibrate their water potential with the regions in the soil where the roots are distributed. The twig isotopic composition of these trees was similar to that of the deepest soil water samples. Therefore, the water potential and isotope values indicate that the deep soil layer was an important plant water source for these nine trees.

The significant relationship between predawn and deep-soil water potentials ($r^2 = 0.46$, $P < 0.01$) at all sites supports the view that the deep soil layer was an important plant water source. In contrast, the relationship between predawn and groundwater potentials was poor ($r^2 = 0.032$, $P = 0.52$).

Rainfall (0.7‰ $\delta^2\text{H}$ and -1.50‰ $\delta^{18}\text{O}$) on the morning of sampling at CF14 was reflected by higher soil water potentials and isotopic composition (Fig. 4) of the surface soils. The remainder of the soil profile was dry and saline (negative matric and osmotic water potentials, respectively) to just above the water table at 1.6-m depth. The $\delta^2\text{H}$ profile (Fig. 4a) shows that there was isotopic enrichment (less negative values) of these shallow soils down to ~ 0.7 -m depth. The CF14c twig water potential and isotope values

were similar to that of the soil sample from 1.6-m depth (Fig. 4). Figure 4a shows that the twig, deep soil water and groundwater samples had similar $\delta^2\text{H}$ (about -30‰) values; however, the groundwater water potential was more negative than the deep soil water or predawn twig values (Fig. 4b), indicating that the groundwater was not available to the tree. The isotope and water potential values indicate that the tree was using water from the deepest soil layer. The isotopic enrichment of the twig water relative to the deep soil water and water potential of the surface soils indicate that the tree may have also used some water from the surface soils.

The profiles from GG8 and GG9 (Table 2) were similar to that shown in Fig. 4, where the soil samples were dry and isotopically enriched down to ~ 1 -m depth. The isotopic composition of the deep soil samples from CF14 and GG9 were similar to the underlying groundwater (about -30‰). However, the deep soil water potentials were less negative than the groundwater and more similar to the predawn water potentials. This again indicates that the deep soil layer was the primary plant water source and that groundwater was not available to these trees. In contrast to CF14 and GG9, soil samples below 1-m depth from GG8 were isotopically enriched, reaching a maximum in the deepest sample (1.4‰ $\delta^2\text{H}$ and 1.96‰ $\delta^{18}\text{O}$). These values were within the range of the observed surface water samples. The match between the soil samples from between 1.0- and 1.6-m depth and the twig isotope and water potential results at GG8 indicate plant water extraction from this depth.

Two soil profiles were taken from site GG5. The profile from near tree GG5a was shallow, moist, relatively fresh (water potentials greater than -1.0 MPa) and isotopically enriched (Fig. 5). Another soil profile taken from GG5a in May 2000 was more saline (soil water potentials from -2.8 to -1.6 MPa; isotope samples were not taken;

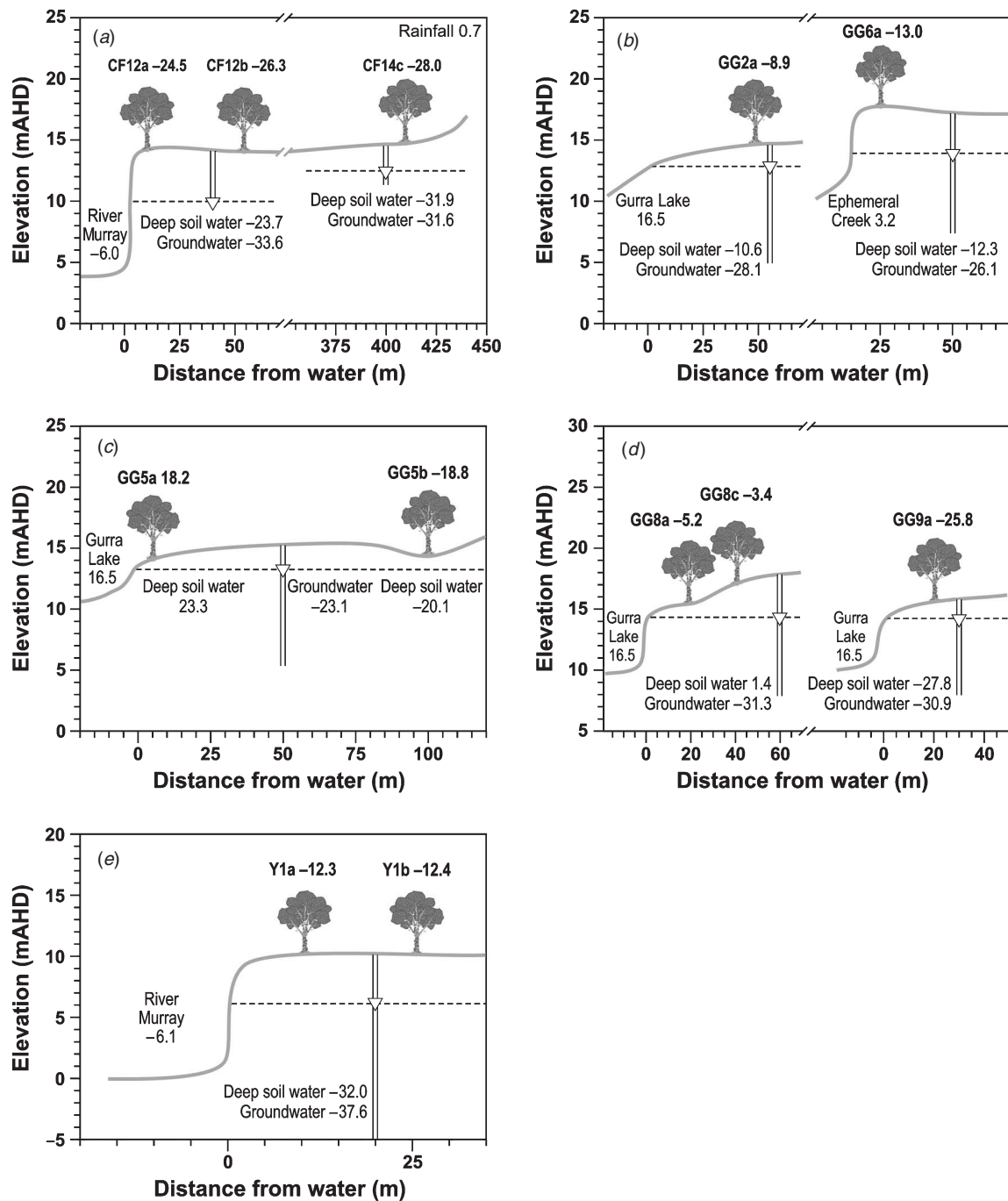


Fig. 3. Transects of sites at Clarks (a) CF12 and CF14; Gurra (b) GG2 and GG6, (c) GG5 and (d) GG8 and GG9; and Yarra (e) Y1 floodplains sampled in February 2001. Tree location is shown relative to the nearest surface water body and to the water table. Approximate local topography is shown for reference. $\delta^2\text{H}$ values (‰) of twig water, surface water, groundwater and deep soil samples are shown. The height of the water table measured in site piezometers is shown with an inverted triangle.

Holland 2002), indicating that salt was removed from, or fresh water added to the profile, between May 2000 and February 2001. The GG5 groundwater potential (-2.59 MPa) was more negative than the GG5a twig water potential (-1.73 MPa). The GG5a twig isotope and predawn water

potential did not match either the soil or groundwater isotope signature or water potential values. However, the GG5a twig isotope values (18.2‰ $\delta^2\text{H}$ and 6.03‰ $\delta^{18}\text{O}$) lay midway between the surface water sample from Gurra Lake (16.5‰ $\delta^2\text{H}$ and 4.44‰ $\delta^{18}\text{O}$) and the soil profile values. This

Table 2. Sample depth, matric, osmotic and total (Ψ) water potentials and isotopic values of soil samples taken from Clarks, Gurra and Yarra floodplains in February 2001

Values highlighted in bold are less than the measured predawn water potential, meaning they are available to the trees

Site	Depth (m)	Water potential (MPa)			$\delta^2\text{H}$ (‰)	$\delta^{18}\text{O}$ (‰)	Site	Depth (m)	Water potential (MPa)			$\delta^2\text{H}$ (‰)	$\delta^{18}\text{O}$ (‰)
		Matric	Osmotic	Ψ					Matric	Osmotic	Ψ		
CF12b	0.1	-3.81	-0.92	-4.73	-	-	GG6a	0.1	-1.42	-5.32	-6.75	-6.6	4.32
	0.3	-1.43	-1.06	-2.50	-29.2	-2.63		0.3	-0.43	-4.40	-4.83	-17.4	0.50
	0.5	-1.70	-1.11	-2.81	-33.3	-3.73		0.5	-0.44	-4.91	-5.35	-19.8	-0.35
	0.7	-1.62	-0.96	-2.58	-33.1	-3.74		0.7	-0.50	-4.57	-5.07	-20.8	-1.05
	0.9	-1.69	-0.91	-2.59	-30.6	-3.62		0.9	-0.44	-4.78	-5.22	-19.4	-0.93
	1.1	-1.95	-0.90	-2.85	-30.4	-3.57		1.1	-0.77	-3.92	-4.69	-19.4	-0.91
	1.6	-0.12	-1.16	-1.28	-27.4	-3.37		1.6	-2.67	-3.71	-6.38	-21.3	-1.44
	2.1	-0.12	-1.12	-1.24	-27.5	-3.28		2.1	-4.37	-3.27	-7.64	-22.2	-1.60
CF14c	2.6	-0.01	-0.66	-0.66	-23.7	-3.09	GG8b	2.6	-0.85	-3.13	-3.98	-19.3	-1.14
	0.1	-0.16	-0.33	-0.49	-3.1	5.45		3.1	-0.02	-2.66	-2.68	-21.8	-0.98
	0.3	-2.84	-0.12	-2.95	-20.8	2.42		3.6	0.00	-1.18	-1.18	-12.3	-0.70
	0.5	-0.22	-2.42	-2.65	-34.8	-1.35		0.1	-43.64	-1.18	-44.82	-	-
	0.7	-0.03	-2.17	-2.20	-45.8	-4.05		0.3	-11.60	-0.74	-12.35	-	-
	0.9	-0.01	-2.58	-2.59	-43.9	-4.16		0.5	-4.83	-0.85	-5.68	-	-
	1.1	-0.01	-2.21	-2.21	-37.4	-3.63		0.7	-2.85	-2.06	-4.92	-19.8	-0.95
	1.6	0.00	-1.63	-1.63	-31.9	-3.42		0.9	-0.59	-2.02	-2.61	-22.1	-1.54
GG2a	0.1	-0.05	-2.46	-2.51	-9.3	-	GG9a	1.1	-0.05	-1.31	-1.36	-17.3	-1.03
	0.3	-0.04	-2.41	-2.45	-17.8	-		1.6	-0.02	-1.37	-1.39	-5.9	0.80
	0.5	-0.03	-3.14	-3.17	-20.4	-0.98		2.1	-0.01	-0.59	-0.59	1.4	1.96
	0.7	-0.04	-1.93	-1.97	-19.9	-1.06		0.1	-0.44	-0.12	-0.55	2.7	4.86
	0.9	-0.01	-1.80	-1.81	-22.6	-1.40		0.3	-1.69	-0.15	-1.84	-2.1	3.85
	1.1	-0.01	-1.97	-1.98	-19.1	-1.55		0.5	-2.96	-1.65	-4.61	-14.8	1.43
	1.6	-0.01	-1.77	-1.78	-20.0	-1.49		0.7	-3.08	-1.70	-4.78	-29.9	-1.65
	2.1	0.00	-1.55	-1.55	-10.6	-0.19		0.9	-1.83	-3.49	-5.32	-30.9	-2.26
GG5a	0.1	-0.02	-0.81	-0.83	23.2	7.36	Y1b	1.1	-0.12	-2.58	-2.71	-26.4	-2.52
	0.3	-0.01	-0.94	-0.94	21.2	6.56		1.6	-0.04	-2.32	-2.36	-26.6	-2.87
	0.5	0.00	-0.90	-0.90	21.5	6.17		2.1	-0.01	-2.12	-2.13	-27.4	-3.22
	0.7	0.00	-0.99	-1.00	21.0	6.28		2.3	-0.01	-2.27	-2.27	-27.8	-3.11
	0.9	0.00	-0.95	-0.95	23.3	6.06		2.4	0.00	-2.31	-2.31	-27.8	-3.34
GG5b	0.1	-8.72	-1.20	-9.92	-	-		0.1	-35.12	-2.06	-37.19	-	-
	0.3	-4.01	-1.58	-5.59	-	-		0.3	-10.69	-1.99	-12.68	-	-
	0.5	-2.56	-1.92	-4.49	-24.3	-1.65		0.5	-6.38	-1.89	-8.27	-	-
	0.7	-0.99	-1.92	-2.91	-22.2	-1.74		0.7	-3.50	-1.95	-5.45	-	-
	0.9	-0.61	-2.23	-2.84	-21.7	-1.59		0.9	-5.59	-2.13	-7.72	-	-
	1.1	-0.07	-1.96	-2.04	-19.9	-1.36		1.1	-1.15	-2.01	-3.16	-22.5	-1.67
	1.6	0.00	-0.98	-0.98	-20.1	-1.63		1.6	-1.24	-1.97	-3.21	-23.8	-2.03
								2.1	-0.05	-2.24	-2.29	-20.7	-1.98
								2.6	-0.04	-2.16	-2.21	-24.6	-2.52

suggests that the tree may have been using a mixture of water from the soil profile and Gurra Lake.

In contrast to the profile from GG5a, the clayey soil profile from GG5b was deeper, drier, more saline (water potentials less than -1.0 MPa) and isotopically depleted relative to the surface water samples (Fig. 5). The GG5 groundwater isotopic composition was similar to that from the deepest soil sample at GG5b, but the groundwater potential was more negative than the deepest soil sample water potential. This suggests that GG5b water extraction was from the soil profile between 1.0- and 1.6-m depths.

Site GG6 is underlain by highly saline groundwater (67 dS m^{-1}) and is adjacent to an ephemeral creek where groundwater seepage was observed. The groundwater sample was more saline and isotopically depleted than the deepest soil sample, whereas the surface water sample was isotopically enriched relative to the twig, soil and

groundwater samples (Fig. 6). The close match between the water potential and isotope values of the GG6a twig and soil water sample from within 0.5 m of the saturated zone indicate that this deep soil layer is the most likely plant water source at this site (Fig. 6).

Similar profiles were observed at CF12 and GG2, where the deepest soil layers were also available to the trees (Table 2). Water potentials in the remainder of the soil profiles were lower than the measured twig predawn water potentials, suggesting they were not available to the trees. Twig and soil isotope and water potential results from CF12 and GG2 also indicate that the deepest soil layers were the primary plant water sources at these sites (Tables 1 and 2). Despite the high water potential of the groundwater at CF12, the close match between the deepest soil sample and twig-isotope values indicate that the trees at CF12 were using soil water from the deepest soil layers, rather than the isotopically depleted

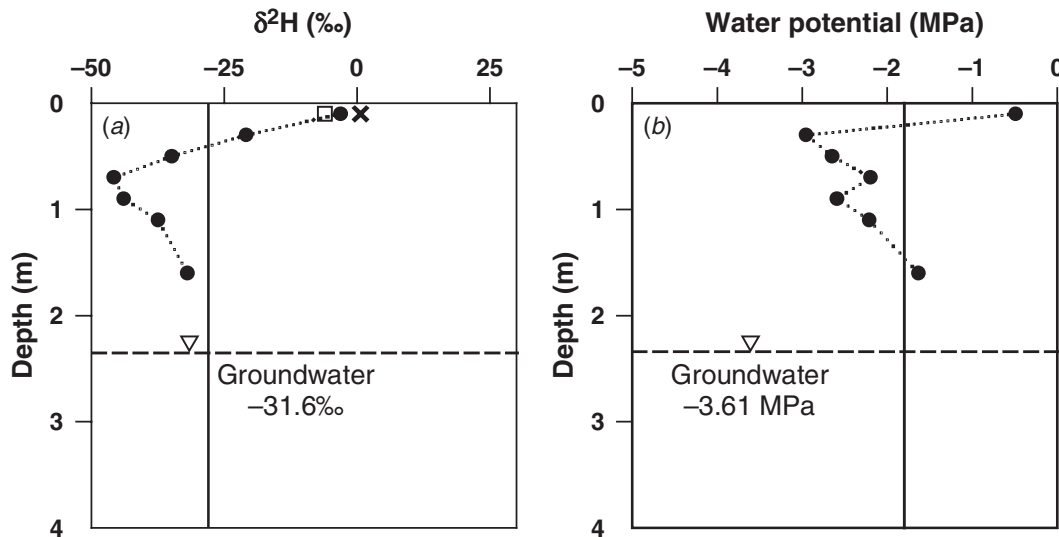


Fig. 4. Site CF14. (a) Plot of soil (●) and groundwater (▽) $\delta^2\text{H}$ (‰) values as a function of depth. The CF14c twig $\delta^2\text{H}$ (‰) value is shown by the vertical line. $\delta^2\text{H}$ (‰) values for rainwater (×) collected on the morning of sampling and the surface water (□) samples collected from the River Murray are shown for reference. (b) Plot of soil (●) and groundwater (▽) water potentials (MPa) as a function of depth. The CF14c twig predawn water potential (MPa) is shown by the vertical line. Groundwater depth below the surface is shown by the horizontal dashed line.

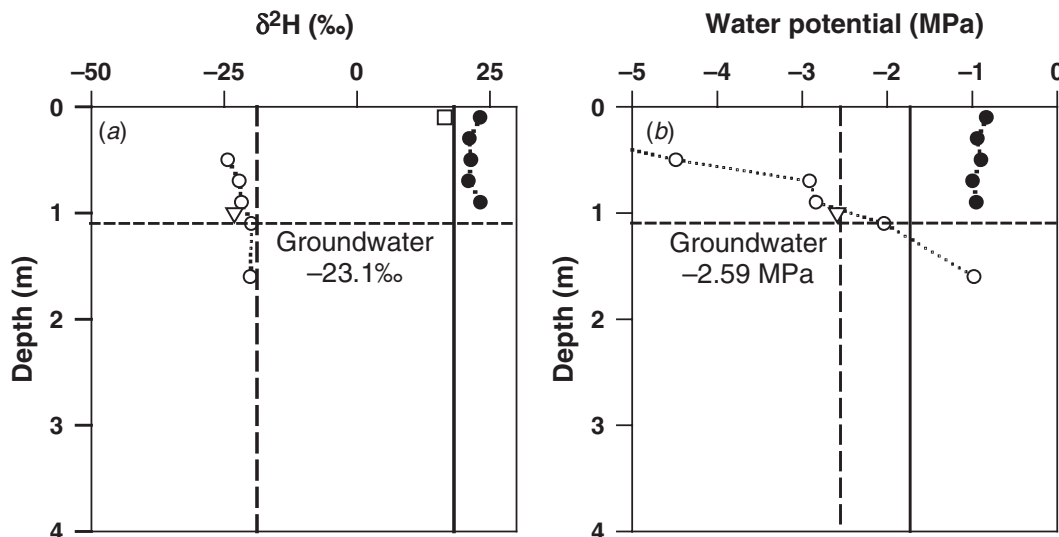


Fig. 5. Site GG5. (a) Plot of GG5a soil (●), GG5b soil (○) and groundwater (▽) $\delta^2\text{H}$ (‰) values as a function of depth. The GG5a (solid line) and GG5b (dashed line) twig $\delta^2\text{H}$ (‰) values are shown by the vertical lines. $\delta^2\text{H}$ (‰) value for the surface water sample (□) collected from the lake is shown for reference. (b) Plot of GG5a soil (●), GG5b (○) soil and groundwater (▽) water potentials (MPa) as a function of depth. The GG5a (solid line) and GG5b (dashed line) twig predawn water potentials (MPa) are shown by the vertical lines. The groundwater depth below the surface is shown by the horizontal dashed line.

groundwater. These sites are relatively high on the floodplain and are characterised by heavy clay profiles and negative soil water potentials, which suggest that they are not frequently inundated by floodwaters.

The isotopic composition of the twig samples from Y1a and Y1b was midway between the River Murray, soil and

groundwater samples. The Y1a predawn water potential was less negative than the soil water potentials, whereas the Y1a twig sample was more isotopically enriched than either the soil or groundwater samples (Fig. 7). The Y1b predawn water potential value coincides with the soil water potential at ~2-m depth, which the soil water potential profile shows

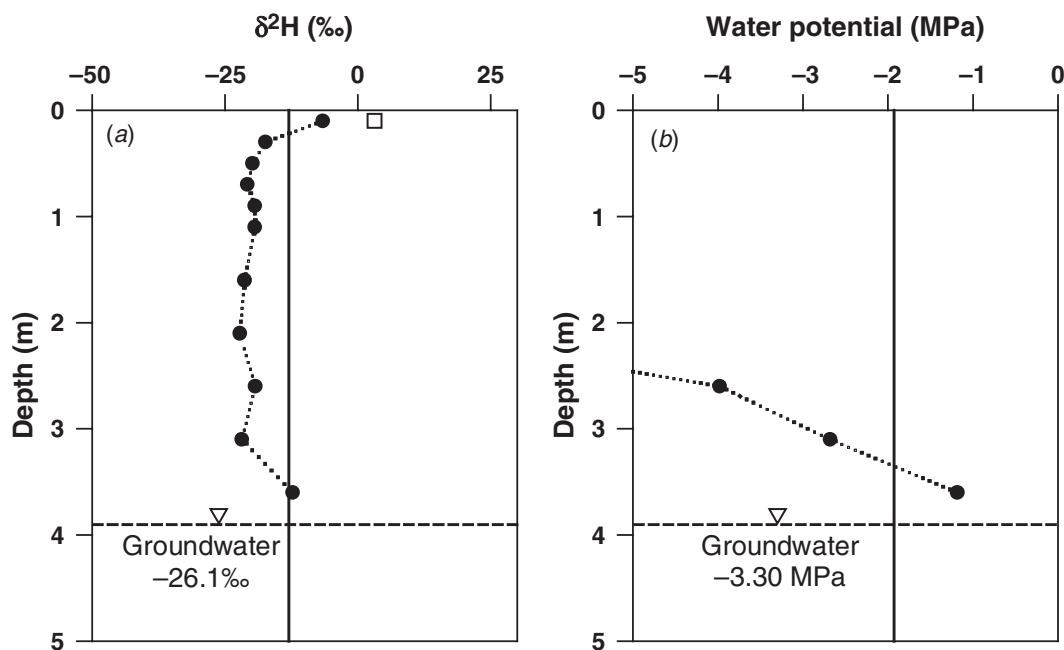


Fig. 6. Site GG6. (a) Plot of soil (●) and groundwater (▽) $\delta^2\text{H}$ (‰) values as a function of depth. The GG6a twig $\delta^2\text{H}$ (‰) value is shown by the vertical line. $\delta^2\text{H}$ (‰) value for the surface water sample (□) collected from the ephemeral creek is shown for reference. (b) Plot of soil (●) and groundwater (▽) water potentials (MPa) as a function of depth. The GG6a twig predawn water potential (MPa) is shown by the vertical line. The groundwater depth below the surface is shown by the horizontal dashed line.

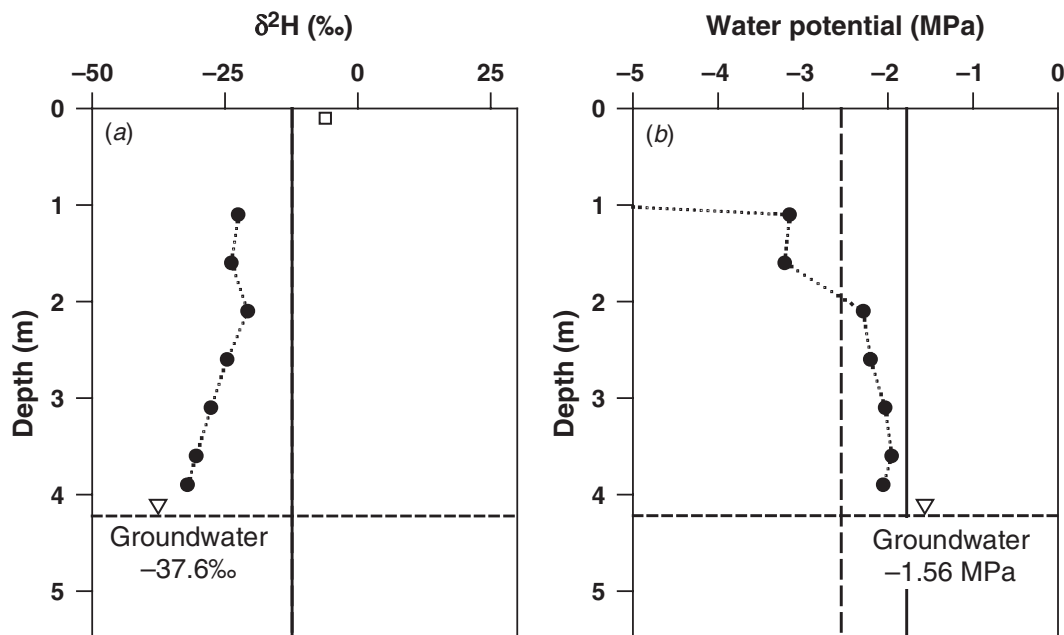


Fig. 7. Site Y1. (a) Plot of Y1 soil (●) and groundwater (▽) $\delta^2\text{H}$ (‰) values as a function of depth. $\delta^2\text{H}$ (‰) values for Y1a (solid line) and Y1b (dashed line) twigs are shown by the overlapping vertical lines. $\delta^2\text{H}$ (‰) value for the surface water sample (□) collected from the River Murray is shown for reference. (b) Plot of Y1 soil (●) and groundwater (▽) water potentials (MPa) as a function of depth. The Y1a (solid line) and Y1b (dashed line) twig predawn water potentials (MPa) are shown by the vertical lines. Groundwater depth below the surface is shown by the horizontal dashed line.

is the top of the drying front above the water table. The presence of this drying front and the match between the Y1b twig and soil water potentials suggest that soil water below ~2-m depth was being used for transpiration. However, the relatively enriched twig water isotopic composition may represent a mixture of isotopically enriched surface water and isotopically depleted soil or groundwater, which results in a twig isotope value midway between the two possible sources. Thus, it is likely that the trees were using a mixture of soil water from below ~2.0-m depth and river water.

Discussion

Plant water sources

Black box trees in this study demonstrated opportunistic water use, using recent rainfall and nearby surface water (where available), in addition to the relatively fresh deep soil water to maintain relatively low plant water stress. This confirms our hypothesis that trees growing above highly saline ($>40 \text{ dS m}^{-1}$) groundwater had access to lower-salinity water sources. The trees were using lower-salinity soil water sources; however, there was no evidence that surface water was being used by the trees at the time of sampling. Opportunistic plant water-use strategies have been demonstrated for trees growing in environments that experience periodic water deficits (e.g. *Eucalyptus*—Sinclair 1980; Landsberg and Wylie 1983; Crombie *et al.* 1988; Thorburn *et al.* 1993a; Mensforth *et al.* 1994; Thorburn and Walker 1993; Jolly and Walker 1996; Akeroyd *et al.* 1998; Zubrinich *et al.* 2000; *Melaleuca*—Mensforth and Walker 1996; *Eucalyptus* and *Acacia*—Thomas and Eamus 2002; *Banksia*—Zencich *et al.* 2002).

The poor relationship between predawn water potentials and groundwater osmotic potentials contrasts with the findings of previous studies where groundwater potentials were related to predawn water potentials (Thorburn *et al.* 1993a; Mensforth *et al.* 1994; Zubrinich *et al.* 2000). However, groundwater samples in these previous studies were taken from the deepest soil samples, rather than site piezometers. The minimum (most saline) osmotic potentials were observed in the deepest soil samples from Zubrinich *et al.* (2000), whereas Thorburn *et al.* (1993a) and Mensforth *et al.* (1994) observed relatively high (greater than -1.5 MPa) osmotic potentials in the groundwater or deepest soil water samples, which is similar to the findings of this study.

Recharge mechanisms

The presence of a layer of deep soil water of low salinity suggests that fresh water was periodically recharged above the deeper, more saline groundwater at most sites. We hypothesised that the layer of deep soil water of low salinity was periodically recharged either (1) vertically through the soil profile or via preferential flow paths by rainfall or flood waters or (2) horizontally by bank recharge from surface water on top of the saline groundwater (Fig. 8). Tree location, water potential and isotopic measurements were used to test this hypothesis and are discussed below.

Vertical infiltration of rain and floodwaters

Floodwaters are known to vertically infiltrate floodplain soil profiles down to 0.5–2.0-m depth on the Chowilla floodplain (Akeroyd *et al.* 1998) and down to 6-m depth in the Barmah–Millewa forest (Bacon *et al.* 1993), depending

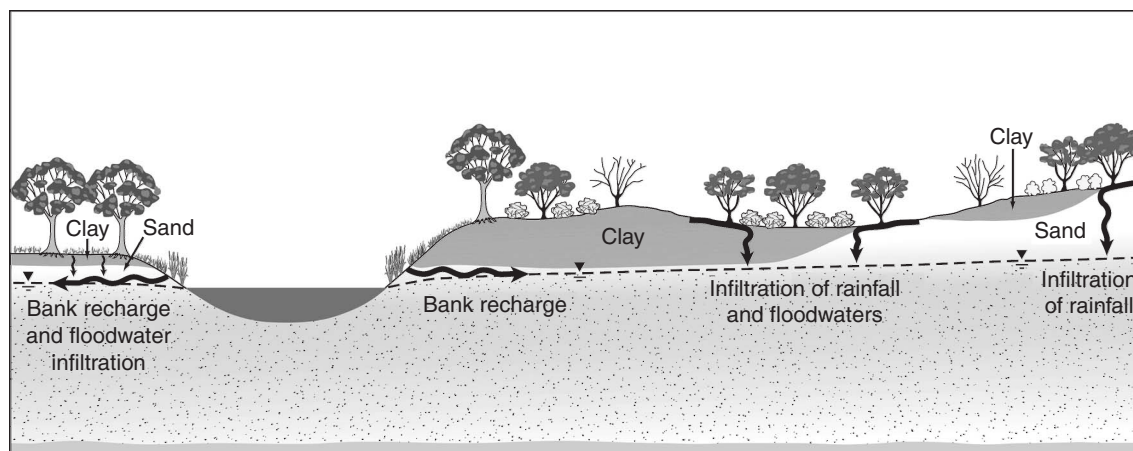


Fig. 8. Conceptual model of deep-soil water-recharge mechanisms that are important for trees growing on the lower River Murray floodplain. Recharge mechanisms were, from left to right, vertical infiltration and horizontal bank recharge on low-lying, sandy river and creek banks; horizontal bank recharge above the groundwater through steep clayey river and creek banks; vertical infiltration of rainfall and floodwaters through preferential flow paths in clayey soils and by diffuse flow through sandy soils where the clay was absent; and vertical infiltration of rainfall through sandy soils at the break of slope.

on soil properties. Recent flooding experiments on lower River Murray floodplains demonstrated that water may infiltrate rapidly to the water table via macropores, particularly beneath tree canopies (Bramley *et al.* 2003).

Vertical infiltration of rain and floodwater to the water table, possibly by preferential flow paths before the swelling, sodic clays dispersed, was inferred at GG5b. The location of GG5b in a topographic depression ~100 m from the nearest surface water in cracking clay soils and the presence of low-salinity, isotopically depleted deep soil water, suggests that isotopically depleted rainfall ($\delta^2\text{H}$ -38‰ ; Simpson and Herczeg 1991) or large floods (e.g. samples taken during floods in December 1992 at Chowilla, $\delta^2\text{H}$ -14.1‰ ; flow 90 GL day^{-1} ; Akeroyd *et al.* 1998) infiltrates to the water table at this site. In contrast, the nearest surface water source was isotopically enriched relative to the deep soil water, indicating that it was not the source of the deep soil water. It is likely that vertical infiltration to the water table is highly spatially and temporally variable owing to the spatial variability of cracking clays and tree-root channels (Hollingsworth 1989; Bacon *et al.* 1993; Bramley *et al.* 2003) and temporal variability of rainfall and floods.

In contrast to Site GG5b, the soil profiles from sites CF14 and GG9 indicated that rainfall had mixed with resident soil water, followed by evaporation of the moist soils down to ~1.0-m depth (Allison *et al.* 1983; Walker *et al.* 1988). Below the isotopic minimum at ~1.0-m depth, the profiles show evidence of mixing with deeper soil water, suggesting that the entire profile is periodically wetted up following heavy rainfall. The results from CF14 contrast with those from CF12, where the surface soil sample taken from CF12 on the same day was too dry for isotopic analysis, whereas the CF14 profile was sandy and saturated by recent rainfall (Table 2). This demonstrates the effect that site and soil characteristics have on rainfall infiltration, meaning that direct infiltration is limited to areas with sandy soils at the break of slope (e.g. CF14, GG8 and GG9) and to areas of topographic depression with preferential flow paths for vertical infiltration (e.g. GG5b).

Horizontal bank recharge by lake, creek and river water

Bank recharge by evaporated lake water to the deep soil layer was observed at sites near Gurra Lake. The deep-soil water samples from GG2, GG5a and GG8 were isotopically enriched and fresh relative to the underlying, saline groundwater (Fig. 3). Sites GG2 and GG5a lie on sandy clay soils, whereas GG8 is on a sandy peninsula between the highland and the lake. All three sites are within 5–50 m of Gurra Lake (Fig. 3). Interpretation of these profiles shows that lake water appears to have recharged the bottom of the soil profile at different times. The isotopic composition of the deepest soil sample from GG2 was comparable with that of floodwaters ($\delta^2\text{H}$ -14.1‰ ; Akeroyd *et al.* 1998), whereas

the isotopic composition of the deepest soil sample from GG8 was intermediate between those of the lake water sample ($\delta^2\text{H}$ 16.5‰) and floodwaters. The upper portion of both profiles (0–0.5-m depth) showed evidence of evaporative enrichment of surface soils, possibly from evaporation of infiltrated rainfall. This indicates that these trees could access water in the upper soil layers, following rainfall, and in the deeper soil layers as this layer dries out over summer. The moist and isotopically enriched soil profile from GG5a indicates that this profile was saturated during the minor flood in December 2000. The isotopic enrichment of the soil profile relative to the lake water sample may represent spatial variability in lake-water isotopic composition, or evaporative enrichment of the shallow soil profile. GG5a was 2–3 km from GG8, where the Gurra Lake surface water sample was taken.

The ephemeral creek also appears to be the source of water to the deep soil water sample at GG6, where the isotopic composition of the deep soil water sample at GG6 ($\delta^2\text{H}$ -12.3‰) lies between River Murray water samples taken in February 2001 ($\delta^2\text{H}$ -6.0‰) and samples taken during floods in December 1992 at Chowilla ($\delta^2\text{H}$ -14.1‰ ; Akeroyd *et al.* 1998). The location of GG6 suggests that floodwater recharge penetrates ~25 m into the clayey creek bank at this site. The reliance of trees at GG6 on rainfall-derived surface soil water before it evaporates, indicated by the dryness and isotopic enrichment of the surface soil samples, and the layer of deep soil water above the saline groundwater means that bank recharge from the ephemeral creek is essential for the trees' survival.

Hydraulic redistribution is an alternative mechanism that could explain the presence of low-salinity, isotopically depleted water in the deep soil layers. The movement of water from areas of high water potential to low water potential, i.e. between different soil layers or from creeks or fresh groundwater into dryer parts of the soil profile, has been demonstrated in riparian environments (Dawson 1993; Dawson and Pate 1996; Burgess *et al.* 1998, 2001; Caldwell *et al.* 1998). However, the isotope results showed that, generally, the deep soil water and surface water isotopic values were dissimilar at the time of sampling. This suggests that infiltration and bank recharge during floods and following rainfall were the more probable recharge mechanisms occurring at these sites.

This study showed that bank recharge was common within ~50 m of the river, ephemeral creeks and backwaters and was an important plant water source in a semi-arid, floodplain environment. This is in agreement with the findings of a previous study that used detailed field measurements to calibrate a small-scale salt-transport model on the Chowilla floodplain (Jolly *et al.* 1998). The salt-transport model and field measurements in the Barmah–Millewa forest showed that groundwater salinity

in the upper parts of the aquifer was reduced within ~40 m of floodplain creeks following floods (Bacon *et al.* 1993; Jolly *et al.* 1998). The current study has shown that bank recharge occurs in many other parts of the floodplain and is an important plant water source for trees growing above highly saline groundwater. This 'bank recharge zone' coincides with the corridor of riparian vegetation typical of many low land floodplain rivers that are underlain by saline groundwater.

Conclusions

This study found that black box trees used water from the deep soil water opportunistically with other water sources, such as rainfall and nearby surface water (where available) to maintain relatively high plant water potentials during periods of water deficit. The isotope and water potential results indicated that the low-salinity deep soil water was recharged in the following two ways:

- (1) vertically by direct infiltration of rainfall (and floodwaters), which was important for trees growing in small depressions and at the break of slope; or
- (2) horizontally by bank recharge from surface water, including evaporated backwaters or lakes (isotopically enriched) and the River Murray or ephemeral creeks (isotopically depleted).

It is unlikely that floodplain management can alter the location and frequency of direct infiltration of rainfall. However, the importance of bank recharge means that alterations to the flow regime typical of many regulated, semi-arid rivers (Jolly 1996) are likely to affect the long-term viability of these riparian vegetation communities. Of particular concern are reductions in the range of river levels associated with both within-channel and overbank flows that reduce the potential for bank recharge to occur. The results of this study also suggest that periodic drying of backwaters or lowering of river levels to lower floodplain groundwater levels, and thus increase bank storage volumes, may increase the effectiveness of bank recharge by minor floods. However, the effectiveness of this approach is difficult to quantify, as the quantification of water fluxes between surface-water and groundwater systems is an area that has been identified as a major challenge requiring further research (Sophocleous 2002).

This study has contributed to a better understanding of the link between plant water sources and recharge from rainfall and surface water. However, although bank recharge may continue to support adult trees, the very dry and saline clayey soil profiles at sites where flooding is infrequent suggest that the recruitment and establishment of saplings at these sites is limited. This has implications for the long-term survival of the riparian vegetation community in the absence of soil profile saturation by heavy rain or floodwater inundation.

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References

- Akeroyd MD, Tyerman SD, Walker GR, Jolly ID (1998) Impact of flooding on the water use of semi-arid riparian eucalypts. *Journal of Hydrology* **206**, 104–117. doi: 10.1016/S0022-1694(98)00092-4
- Allison GB, Barnes CJ, Hughes MW (1983) The distribution of deuterium and ^{18}O in dry soils. 2. Experimental. *Journal of Hydrology* **64**, 377–397. doi: 10.1016/0022-1694(83)90078-1
- Bacon PE, Stone C, Binns DL, Leslie DJ, Edwards DW (1993) Relationships between water availability and *Eucalyptus camaldulensis* growth in a riparian forest. *Journal of Hydrology* **150**, 541–561. doi: 10.1016/0022-1694(93)90125-S
- Barnes CJ, Allison GB (1983) The distribution of deuterium and oxygen-18 in dry soil. 1. Theory. *Journal of Hydrology* **60**, 141–156. doi: 10.1016/0022-1694(83)90018-5
- Bramley H, Hutson J, Tyerman SD (2003) Floodwater infiltration through root channels on a sodic clay floodplain and the influence on a local tree species *Eucalyptus largiflorens*. *Plant and Soil* **253**, 275–286. doi: 10.1023/A:1024531325281
- Bureau of Meteorology (2001) 'Climate averages for Loxton Research Centre (024024) and Waikerie (024018).' <http://www.bom.gov.au/climate/averages/> [accessed 15 March 2006].
- Burgess SSO, Adams MA, Turner NC, Ong KC (1998) The redistribution of soil water by tree root systems. *Oecologia* **115**, 306–311. doi: 10.1007/s004420050521
- Burgess SSO, Adams MA, Turner NC, White DA, Ong CK (2001) Tree roots: conduits for deep recharge of soil water. *Oecologia* **126**, 158–165. doi: 10.1007/s004420000501
- Caldwell MM, Dawson TE, Richards JH (1998) Hydraulic lift: consequences of water efflux from the roots of plants. *Oecologia* **113**, 151–161. doi: 10.1007/s004420050363
- Chimner RA, Cooper DJ (2004) Using stable oxygen isotopes to quantify the water source used for transpiration by native shrubs in the San Luis Valley, Colorado USA. *Plant and Soil* **260**, 225–236. doi: 10.1023/B:PLSO.0000030190.70085.e9
- Craig AE, Walker KF, Boulton AJ (1991) Effects of edaphic factors and flood frequency on the abundance of lignum (*Muehlenbeckia florulenta* Meissner) (Polygonaceae) on the River Murray floodplain, South Australia. *Australian Journal of Botany* **39**, 431–443. doi: 10.1071/BT9910431
- Crombie DS, Tippet JT, Hill TC (1988) Dawn water potential and root depths of trees and understorey species in south-western Australia. *Australian Journal of Botany* **36**, 621–631. doi: 10.1071/BT9880621

- Dawson TE (1993) Hydraulic lift and water use by plants: implications for water balance, performance and plant-plant interactions. *Oecologia* **95**, 565–574.
- Dawson TE, Ehleringer JR (1991) Streamside trees that do not use stream water. *Nature* **350**, 335–337. doi: 10.1038/350335a0
- Dawson TE, Pate JS (1996) Seasonal water uptake and movement in root systems of Australian phreatophytic plants of dimorphic root morphology: a stable isotope investigation. *Oecologia* **107**, 13–20. doi: 10.1007/BF00582230
- DEH (2004) 'Floodplain impacts project native vegetation data.' (Department for Environment and Heritage: Adelaide)
- Dighton JC, Leaney FW, Herczeg AL, Allison GB, Hughes MW (1997) A rapid and robust method for the preparation of isotopically unaltered hydrogen gas from water for stable isotope mass spectrometry. In '6th Australian–New Zealand environmental isotope conference', Wellington, New Zealand.
- Eagleson PS (1982) Ecological optimality in water-limited natural soil-vegetation systems. 1. Theory and hypothesis. *Water Resources Research* **18**, 325–340.
- Flanagan LB, Ehleringer JR, Marshall JD (1992) Differential uptake of summer precipitation among co-occurring trees and shrubs in a pinyon-juniper woodland. *Plant, Cell & Environment* **15**, 831–836.
- Gardner WR (1960) Dynamic aspects of water availability to plants. *Soil Science* **89**, 63–73.
- Gat JR (1981) Isotopic fractionation. Chapter 3. In 'Stable isotope hydrology: deuterium and oxygen-18 in the water cycle'. (Eds JR Gat, R Gonfiantini) pp. 21–33. (International Atomic Energy Agency: Vienna)
- Greacen EL, Walker GR, Cook PG (1989) Evaluation of the filter paper method for measuring soil water suction. In 'International meeting at University of Utah on measurement of soil and plant water status'. pp. 137–145.
- Hatton TJ, Evans R (1998) 'Dependence of ecosystems on groundwater and its significance to Australia.' LWRRDC Occasional Paper 12/98. (LWRRDC: Canberra)
- Herczeg AL, Dogramaci SS, Leaney FW (2001) Origin of dissolved salts in a large, semi-arid groundwater system: Murray Basin, Australia. *Marine and Freshwater Research* **52**, 41–52. doi: 10.1071/MF00040
- Holland KL (2002) Blackbox (*Eucalyptus largiflorens*) tree health and groundwater discharge on the lower River Murray floodplain: linking spatial and temporal scales. PhD Thesis, Flinders University of South Australia, Adelaide.
- Hollingsworth ID (1989) A reconnaissance soil survey of the Chowilla Anabranch System of the River Murray in South Australia and New South Wales. Murray–Darling Basin Commission by the Loxton Research Centre, South Australian Department of Agriculture, Adelaide.
- Jolly ID (1996) The effects of river management on the hydrology and hydroecology of arid and semi-arid floodplains. In 'Floodplain processes'. (Eds MG Anderson, DE Walling, PD Bates) pp. 577–609. (John Wiley & Sons Ltd: Chichester, New York)
- Jolly ID, Walker GR (1996) Is the field water use of *Eucalyptus largiflorens* F.Muell. affected by short-term flooding? *Australian Journal of Ecology* **21**, 173–183.
- Jolly ID, Narayan KA, Armstrong D, Walker GR (1998) The impact of flooding on modelling salt transport processes to streams. *Environmental Modelling & Software* **13**, 87–104. doi: 10.1016/S1364-8152(98)00003-6
- Kozłowski TT (1968) 'Water deficits and plant growth. Vol. 1.' (Academic Press: London and New York)
- Landsberg J, Wylie FR (1983) Water stress, leaf nutrients and defoliation: a model of dieback of rural eucalypts. *Australian Journal of Ecology* **8**, 27–41.
- MDBC (2001) 'River Murray flows to South Australia.' <http://www.mdbc.gov.au> [accessed 15 March 2006].
- MDBMC (1999) The salinity audit of the Murray–Darling Basin. A 100-year perspective. (Murray–Darling Basin Ministerial Council: Canberra)
- Mensforth LJ, Walker GR (1996) Root dynamics of *Melaleuca halmaturorum* in response to fluctuating saline groundwater. *Plant and Soil* **184**, 75–84. doi: 10.1007/BF00029276
- Mensforth LJ, Thorburn PJ, Tyerman SD, Walker GR (1994) Sources of water used by riparian *Eucalyptus camaldulensis* overlying highly saline groundwater. *Oecologia* **100**, 21–28. doi: 10.1007/BF00317126
- Nemani RR, Running SW (1989) Testing a theoretical climate–soil–leaf area hydrologic equilibrium of forests using satellite data and ecosystem simulation. *Agricultural and Forest Meteorology* **44**, 245–260. doi: 10.1016/0168-1923(89)90020-8
- O'Malley C, Sheldon F (1990) 'Chowilla floodplain biological study.' (Nature Conservation Society of South Australia: Adelaide)
- Revesz K, Woods PH (1990) A method to extract soil water for stable isotope analysis. *Journal of Hydrology* **115**, 397–406. doi: 10.1016/0022-1694(90)90217-L
- Ritchie GA, Hinckley TM (1975) The pressure chamber as an instrument for ecological research. *Advances in Ecological Research* **9**, 165–254.
- Scholander PF, Hammel HT, Bradstreet ED, Hemmingsen EA (1965) Sap pressure in vascular plants. *Science* **148**, 339–346.
- Simpson HJ, Herczeg AL (1991) Stable isotopes as an indicator of evaporation in the River Murray, Australia. *Water Resources Research* **27**, 1925–1935. doi: 10.1029/91WR00941
- Sinclair R (1980) Water potential and stomatal conductance of three *Eucalyptus* species in the Mount Lofty Ranges, South Australia: Responses to summer drought. *Australian Journal of Botany* **28**, 499–510. doi: 10.1071/BT9800499
- Slavich PG, Walker GR, Jolly ID, Hatton TJ, Dawes WRJ (1999) Dynamics of *Eucalyptus largiflorens* growth and water use in response to modified watertable and flooding regimes on a saline floodplain. *Agricultural Water Management* **39**, 245–264. doi: 10.1016/S0378-3774(98)00081-X
- Socki RA, Karlsson HR, Gibson EK (1992) Extraction technique for the determination of oxygen-18 in water using pre-evacuated glass vials. *Analytical Chemistry* **64**, 829–831. doi: 10.1021/ac00031a026
- Sophocleous M (2002) Interactions between groundwater and surface water: the state of the science. *Hydrogeology Journal* **10**, 52–67. doi: 10.1007/s10040-001-0170-8
- Taras MJ, Greenberg AE, Hoak RD, Rand MC (1975) 'Standard methods for the examination of water and wastewater.' (American Public Health Association: Washington, DC)
- Thomas DS, Eamus D (2002) Seasonal patterns of xylem sap pH, xylem abscisic acid concentration, leaf water potential and stomatal conductance of six evergreen and deciduous Australian savanna tree species. *Australian Journal of Botany* **50**, 229–236. doi: 10.1071/BT01045
- Thorburn PJ, Walker GR (1993) The source of water transpired by *Eucalyptus camaldulensis*: soil, groundwater or streams? In 'Stable isotopes and plant carbon–water relations'. (Eds JR Ehleringer, AE Hall, GD Farquhar) pp. 511–527. (Academic Press: San Diego, CA).
- Thorburn PJ, Hatton TJ, Walker GR (1993a) Combining measurements of transpiration and stable isotopes of water to determine groundwater discharge from forests. *Journal of Hydrology* **150**, 563–587. doi: 10.1016/0022-1694(93)90126-T
- Thorburn PJ, Walker GR, Brunel J-P (1993b) Extraction of water from *Eucalyptus* trees for analysis of deuterium and oxygen-18: laboratory and field techniques. *Plant, Cell & Environment* **16**, 269–277.

- Turner NC (1988) Measurement of plant water status by the pressure chamber technique. *Irrigation Science* **9**, 289–308. doi: 10.1007/BF00296704
- Walker GR, Hughes MW, Allison GB, Barnes CJ (1988) The movement of isotopes of water during evaporation from a bare soil surface. *Journal of Hydrology* **97**, 181–197. doi: 10.1016/0022-1694(88)90114-X
- Weast RC, Astle MJ (1980) Table 71 D-262. In 'CRC handbook of chemistry and physics'. (CRC Press Inc.: Boca Raton, FL)
- White JWC, Cook ER, Lawrence JR, Broeker WS (1985) The D/H ratios of sap in trees: Implications for water sources and tree ring D/H ratios. *Geochimica et Cosmochimica Acta* **49**, 237–246. doi: 10.1016/0016-7037(85)90207-8
- Zencich SJ, Froend RH, Turner JV, Gailitis V (2002) Influence of groundwater depth on the seasonal sources of water accessed by *Banksia* tree species on a shallow, sand coastal aquifer. *Oecologia* **131**, 8–19. doi: 10.1007/s00442-001-0855-7
- Zubrinich TM, Loveys BR, Gallasch S, Seekamp JV, Tyerman SD (2000) Tolerance of salinised floodplain conditions in a naturally occurring *Eucalyptus* hybrid related to lowered plant water potential. *Tree Physiology* **20**, 953–963.

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